Department of Artificial Intelligence

Indian Institute of Technology Hyderabad

Final Examination

AI2200: Concentration Inequalities

July-August 2026

Student Details

Name ______________________________

Roll Number ______________________________

02 September 2026 | **15 Marks** | **Part 01**

Instructions

1. Fill in your name and roll number in the space provided above.
2. This part is for 15 marks. The complete examination is for 30 marks and counts towards 50% of the total grade.
3. Show all your working clearly. Intermediate logical steps may receive partial credit.
4. Plagiarism will not be entertained. If you are caught cheating during the examination, your answer script will not be evaluated.

1 Useful Results

1. A random variable X is said to be sub-Gaussian if it satisfies any of the following equivalent conditions:

   (a) There exists $K_1 > 0$ such that
   $$\mathbb{P}(\{|X| \geq \varepsilon\}) \leq 2\exp\left(-\frac{\varepsilon^2}{K_1^2}\right) \qquad \forall\, \varepsilon > 0.$$

   (b) There exists $K_2 > 0$ such that
   $$\Big(\mathbb{E}[|X|^p]\Big)^{1/p} \leq K_2\sqrt{p} \qquad \forall\, p \geq 1.$$

   (c) There exists $K_3 > 0$ such that
   $$\mathbb{E}[e^{t^2X^2}] \leq e^{t^2K_3^2} \qquad \forall\, t \in \left[-\frac{1}{K_3}, \frac{1}{K_3}\right].$$

   (d) There exists $K_4 > 0$ such that
   $$\mathbb{E}[e^{X^2/K_4^2}] \leq 2.$$

   Additionally, if $\mathbb{E}[X] = 0$:

   (e) There exists $K_5 > 0$ such that
   $$\mathbb{E}[e^{tX}] \leq e^{t^2\,K_5^2} \qquad \forall\, t \in \mathbb{R}.$$

   The best constants $K_1^\star, \ldots, K_5^\star$ satisfy the property
   $$c_{ij} \leq \frac{K_i^\star}{K_j^\star} \leq C_{ij} \qquad \forall\, i \neq j,$$
   where c_{ij} and C_{ij} are universal constants.
   One may replace X with $X - \mathbb{E}[X]$ in all of the above conditions.

2. The sub-Gaussian norm of a sub-Gaussian random variable X is defined as
   $$\|X\|_{\psi_2} = \inf\left\{K > 0 : \ \mathbb{E}[e^{X^2/K^2}] \leq 2.\right\}$$

3. A random variable X is said to be σ-sub-Gaussian if
   $$\mathbb{E}[e^{t(X-\mathbb{E}[X])}] \leq \exp\left(\frac{t^2\sigma^2}{2}\right) \qquad \forall\, t \in \mathbb{R}.$$

4. A random variable X that, with probability 1, takes values in a bounded interval $[a, b]$ for some $< a < b$, is $\left(\frac{b-a}{2}\right)$-sub-Gaussian.

5. A function $g : \mathbb{R}^n \to \mathbb{R}$ is said to satisfy the bounded differences property (BDP) if there exist constants $c_1, \ldots, c_n$ such that
   $$\Big|g(x_1, \ldots, x_{i-1}, x, x_{i+1}, \ldots, x_n) - g(x_1, \ldots, x_{i-1}, y, x_{i+1}, \ldots, x_n)\Big| \leq c_i$$
   for all i and for all $x_1, \ldots, x_{n-1}, x, y \in \mathbb{R}$.

Question 1 (2+3 Marks)

Given $n \in \mathbb{N}$ and events $A_1, A_2, \ldots, A_n$, the union bound states that

$$\mathbb{P}\left(\bigcup_{k=1}^{n} A_k\right) \leq \sum_{k=1}^{n} \mathbb{P}(A_k).$$

(a) Prove the union bound via Markov's inequality. Clearly indicate the non-negative random variable to which you apply Markov's inequality.

(b) A direct proof of the union bound is as below:
Define the sequence $B_1, \ldots, B_n$ as

$$B_1 = A_1, \quad B_2 = A_2 \setminus A_1, \quad B_3 = A_3 \setminus (A_1 \cup A_2), \quad \ldots, \quad B_n = A_n \setminus \left(\bigcup_{k=1}^{n-1} A_k\right).$$

Then,

$$\mathbb{P}\left(\bigcup_{k=1}^{n} A_k\right) = \mathbb{P}\left(\bigcup_{k=1}^{n} B_k\right) = \sum_{k=1}^{n} \mathbb{P}(B_k) \leq \sum_{k=1}^{n} \mathbb{P}(A_k).$$

Identify the necessary and sufficient conditions for the union bound to hold with equality.

Note: Do not attempt to derive these conditions based on the condition for equality in Markov's inequality. Use your knowledge from the probability course for deriving the necessary and sufficient conditions based on the proof given above.

Question 2 (3+2 Marks)

Fix $n \in \mathbb{N}$. Let $X_1, \ldots, X_n$ be independent and sub-Gaussian with sub-Gaussian norms

$$\|X_i\|_{\psi_2} = K_i, \qquad i \in \{1, \ldots, n\}.$$

(a) Show that $X = \max\{X_1, \ldots, X_n\}$ is sub-Gaussian.

(b) Derive an upper bound on the sub-Gaussian norm of X in terms of $K_1, \ldots, K_n$. Clearly indicate the place where you use the independence assumption.

Hint: Think of how you can upper bound

$$\mathbb{P}\left(\{|\max\{X_1, \ldots, X_n\}| > \varepsilon\}\right)$$

for any $\varepsilon > 0$.

Question 3 (1+2+2 Marks)

Fix $\alpha \in (0,1)$. Suppose that X_1 and X_2 are sub-Gaussian with sub-Gaussian norms K_1 and K_2, with $0 < K_1 < K_2$. Let $B \sim \mathrm{Ber}(\alpha)$ be independent of X_1 and X_2, and let X be the random variable

$$X(\omega) \quad = \quad \begin{cases} X_1(\omega), & \quad B(\omega) = 1, \\ X_2(\omega), & \quad B(\omega) = 0, \end{cases}$$

for each $\omega \in \Omega$.

(a) If X_1 has PDF f_1 and X_2 has PDF f_2, derive an expression for the PDF f of X.

(b) Prove that X is sub-Gaussian, and give an upper bound on $\|X\|_{\psi_2}$.

(c) Let $\mathbb{E}[X] = \mu$, and let $Y_1, \ldots, Y_n \overset{\text{IID}}{\sim} f$, where f is the PDF of X. Given $\varepsilon > 0$ and $\delta \in (0,1)$, show that

$$n \quad \geq \quad \frac{cK_2^2}{\varepsilon^2} \log\left(\frac{2}{\delta}\right) \qquad \Longrightarrow \qquad \mathbb{P}\left(\left|\frac{1}{n}\sum_{k=1}^{n} Y_k \; - \; \mu\right| \; > \; \varepsilon\right) \quad \leq \quad \delta,$$

where c is a universal constant.

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2 Useful Results

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   $$\mathbb{P}(\{|X| \geq \varepsilon\}) \leq 2\exp\left(-\frac{\varepsilon^2}{K_1^2}\right) \qquad \forall\, \varepsilon > 0.$$

   (b) There exists $K_2 > 0$ such that
   $$\Big(\mathbb{E}[|X|^p]\Big)^{1/p} \leq K_2\sqrt{p} \qquad \forall\, p \geq 1.$$

   (c) There exists $K_3 > 0$ such that
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   (d) There exists $K_4 > 0$ such that
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   One may replace X with $X - \mathbb{E}[X]$ in all of the above conditions.

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   $$\|X\|_{\psi_2} = \inf\left\{K > 0 : \ \mathbb{E}[e^{X^2/K^2}] \leq 2.\right\}$$

3. A random variable X is said to be σ-sub-Gaussian if
   $$\mathbb{E}[e^{t(X-\mathbb{E}[X])}] \leq \exp\left(\frac{t^2\sigma^2}{2}\right) \qquad \forall\, t \in \mathbb{R}.$$

4. A random variable X that, with probability 1, takes values in a bounded interval $[a, b]$ for some $< a < b$, is $\left(\frac{b-a}{2}\right)$-sub-Gaussian.

5. A function $g : \mathbb{R}^n \to \mathbb{R}$ is said to satisfy the bounded differences property (BDP) if there exist constants $c_1, \ldots, c_n$ such that
   $$\Big|g(x_1, \ldots, x_{i-1}, x, x_{i+1}, \ldots, x_n) - g(x_1, \ldots, x_{i-1}, y, x_{i+1}, \ldots, x_n)\Big| \leq c_i$$
   for all i and for all $x_1, \ldots, x_{n-1}, x, y \in \mathbb{R}$.

Question 1 (3 Marks)

Suppose that X_1 and X_2 are sub-Gaussian (not necessarily independent), with sub-Gaussian norms ν_1 and ν_2 respectively. Show that

$$\|X_1 + X_2\|_{\psi_2} \leq \sqrt{2\left(\nu_1^2 + \nu_2^2\right)}.$$

Hint: Using the relation $(a+b)^2 \leq 2(a^2+b^2)$, we get

$$\mathbb{E}\left[\exp\left(\frac{(X_1+X_2)^2}{2(\nu_1^2+\nu_2^2)}\right)\right] \leq \mathbb{E}\left[\exp\left(\frac{X_1^2+X_2^2}{\nu_1^2+\nu_2^2}\right)\right].$$

Proceed with the quantity on the right-hand side of the above inequality.

Question 2 (2+2+3 Marks)

Let $X_1, X_2, \ldots$ be a sequence of random variables (not necessarily mutually independent). Suppose that X_k is sub-Gaussian with sub-Gaussian norm ν_k for each $k \in \mathbb{N}$. Let

$$Z \;:=\; \sup_{k\in\mathbb{N}} \frac{|X_k|}{\sqrt{1+\log k}}, \qquad \nu \;:=\; \sup_{k\in\mathbb{N}} \nu_k.$$

Assume that $\nu < +\infty$.

(a) Show that for any $\varepsilon > 0$,

$$\mathbb{P}\left(\{Z \geq \varepsilon\}\right) \;\leq\; 2\exp(-a)\sum_{k\in\mathbb{N}} \frac{1}{k^a}, \qquad a = \frac{c_0\varepsilon^2}{\nu^2},$$

where c_0 is a universal constant. Clearly justify every step in your calculations.

(b) Using the relation

$$\sum_{k\in\mathbb{N}} \frac{1}{k^a} \;\leq\; 1 + \int_1^\infty \frac{1}{x^a}\,\mathrm{d}x,$$

show that

$$\sum_{k\in\mathbb{N}} \frac{1}{k^a} \;\leq\; 2 \qquad \text{whenever } a \geq 2.$$

Hence show that

$$\mathbb{P}(\{Z \geq \varepsilon\}) \;\leq\; 4\exp(-a) \qquad \text{whenever} \quad \varepsilon \geq \nu\sqrt{\frac{2}{c_0}}.$$

(c) Using the relation

$$\mathbb{E}[Z] \;=\; \int_0^\infty \mathbb{P}(\{Z \geq \varepsilon\})\,\mathrm{d}\varepsilon,$$

split the above integral at $\varepsilon_0 = \nu\sqrt{\frac{2}{c_0}}$ and show that

$$\mathbb{E}[Z] \;\leq\; C\nu$$

for some universal constant $C > 0$.

Question 3 (3+2 Marks)

Given any $A \subseteq \mathbb{R}$, let a binary classifier g_A be defined via

$$g_A : \mathbb{R} \to \{0,1\}, \qquad g_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Furthermore, let $\mathcal{G}$ be the uncountably infinite collection of classifiers given by

$$\mathcal{G} = \{g_A : A \subseteq \mathbb{R}\}.$$

Consider n IID data points $(X_1, Y_1), \ldots, (X_n, Y_n) \in \mathbb{R} \times \{0,1\}$ sampled from an unknown distribution $P_{X,Y}$. For any $A \subseteq \mathbb{R}$, let

$$R(g_A) \;\; := \;\; \mathbb{P}_{(X,Y) \sim P_{X,Y}}\Big(\Big\{g_A(X) \neq Y\Big\}\Big), \qquad \widehat{R}_n(g_A) \;\; := \;\; \frac{1}{n} \sum_{k=1}^{n} \mathbf{1}_{\{g_A(X_k) \,\neq\, Y_k\}}.$$

Let $\Phi : (\mathbb{R} \times \{0,1\})^n \to \mathbb{R}$ be defined as

$$\Phi\Big((X_1, Y_1) \ldots, (X_n, Y_n)\Big) \;\; := \;\; \sup_{A \subseteq \mathbb{R}} \Big[R(g_A) - \widehat{R}_n(g_A)\Big].$$

(a) Does Φ satisfy the bounded differences property (BDP)? If so, identify the constants $c_1, \ldots, c_n$ appearing in the BDP relation.

(b) Show that

$$0 \;\; \leq \;\; \Phi\Big((x_1, y_1), \ldots, (x_n, y_n)\Big) \;\; \leq \;\; 1 \qquad \forall \; (x_1, y_1), \ldots, (x_n, y_n) \in \mathbb{R} \times \{0,1\}.$$